PS2.R

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```r
# ---
# title: "Problem Set 2: The LaCour-Green Study"
# author: "Andrea Moreno"
# date: "2026-03-05"
# output: pdf_document
# ---
# GitHub Repository: "https://github.com/andrealmoreno/gov51-ps2.git"t 1: The origin
#Load data
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.2     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
getwd()
```

```
## [1] "/Users/andreamoreno/Desktop/gov51-ps2/gov51-ps2"
```

```r
gay <- read_csv("data/raw/gay.csv")
```

```
## Rows: 69592 Columns: 4
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): treatment
## dbl (3): study, wave, ssm
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
gay_reshaped <- read_csv("data/raw/gayreshaped.csv")
```

```
## Rows: 11948 Columns: 6
```

```
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (1): treatment
## dbl (5): study, therm1, therm2, therm3, therm4
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
ccap <- read_csv("data/raw/ccap2012.csv")
```

```
## New names:
## Rows: 43998 Columns: 3
## -- Column specification
## -------------------------------------------------------- Delimiter: "," dbl
## (3): ...1, caseid, gaytherm
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...1`
```

```r
head(gay)
```

```
## # A tibble: 6 x 4
##   study treatment   wave   ssm
##   <dbl> <chr>      <dbl> <dbl>
## 1     1 No Contact     3     5
## 2     1 No Contact     4     5
## 3     1 No Contact     1     5
## 4     1 No Contact     6     5
## 5     1 No Contact     2     5
## 6     1 No Contact     7     5
```

```r
#Q1.1
# 1. How many total observations?
nrow(gay)
```

```
## [1] 69592
```

```r
#ANSWER There are 69,592 total observations in gay.csv.

# 2. How many unique values does 'study' take?
# n_distinct() is a handy tidyverse function for this
n_distinct(gay$study)
```

```
## [1] 2
```

```r
# ANSWER: The 'study' variable takes 2 unique values.

# 3. List all treatment conditions
# unique() will show you every distinct label in that column
unique(gay$treatment)
```

```
## [1] "No Contact"
## [2] "Recycling Script by Gay Canvasser"
## [3] "Same-Sex Marriage Script by Gay Canvasser"
## [4] "Recycling Script by Straight Canvasser"
## [5] "Same-Sex Marriage Script by Straight Canvasser"
```

```r
# ANSWER: The treatment conditions are:
# (1) No Contact
# (2) Recycling Script by Gay Canvasser
# (3) Same-Sex Marriage Script by Gay Canvasser
# (4) Recycling Script by Straight Canvasser
# (5) Same-Sex Marriage Script by Straight Canvasser

#Q1.2:

# 1. Filter to Study 1 and Wave 1 only
study1_wave1 <- gay %>%
  filter(study == 1, wave == 1)

# 2. Calculate the mean of ssm for each treatment group
q1_2_results <- study1_wave1 %>%
  group_by(treatment) %>%
  summarize(mean_ssm = mean(ssm, na.rm = TRUE))

# 3. View the results
print(q1_2_results)
```

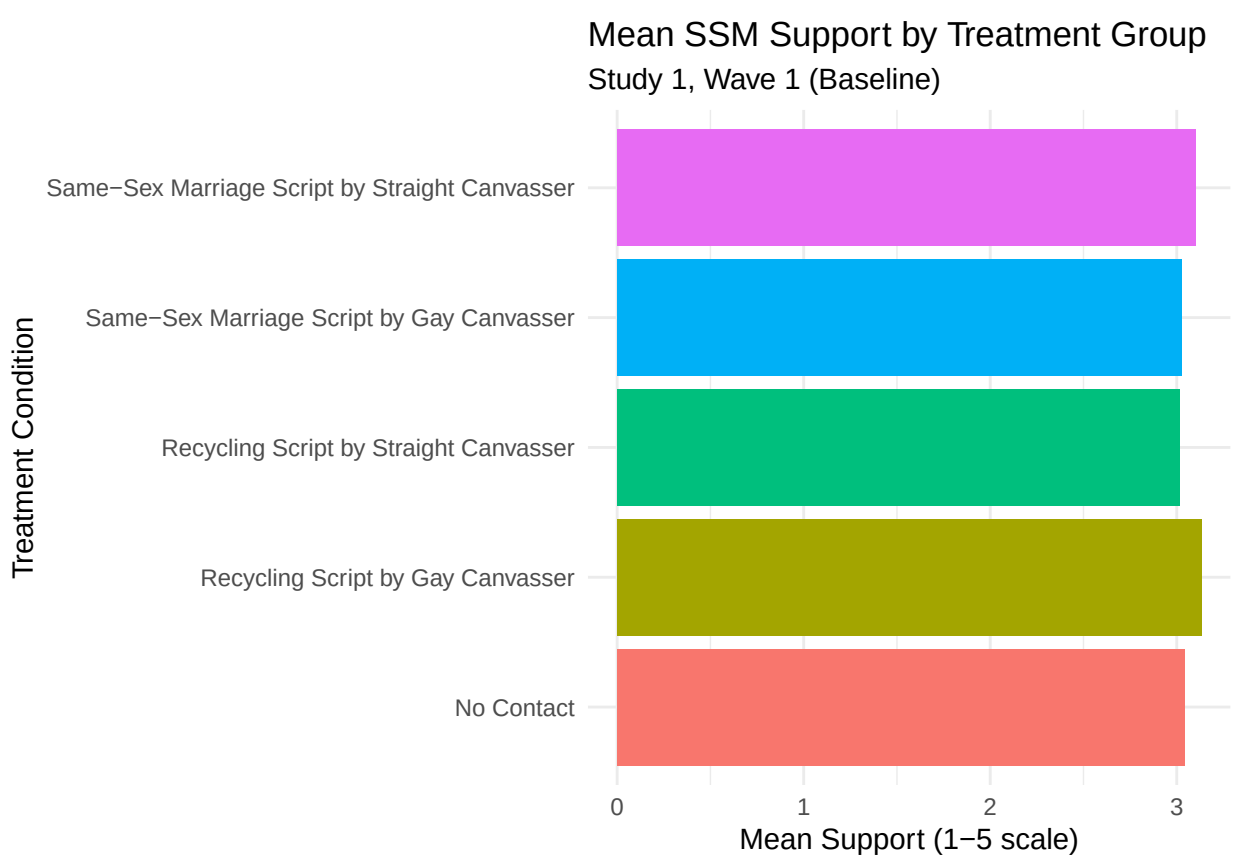
```
## # A tibble: 5 x 2
##   treatment                                       mean_ssm
##   <chr>                                              <dbl>
## 1 No Contact                                          3.04
## 2 Recycling Script by Gay Canvasser                   3.13
## 3 Recycling Script by Straight Canvasser              3.01
## 4 Same-Sex Marriage Script by Gay Canvasser           3.03
## 5 Same-Sex Marriage Script by Straight Canvasser      3.10
```

```r
# ANSWER: In Study 1, Wave 1, the "Recycling Script by Gay Canvasser" group
# shows the highest average support with a mean ssm of 3.13.
# However, all groups are very similar (between 3.01 and 3.13),
# which suggests the randomization worked at the baseline.

#Q1.3

# Calculate the standard deviation of ssm for each treatment group
q1_3_results <- study1_wave1 %>%
  group_by(treatment) %>%
  summarize(sd_ssm = sd(ssm, na.rm = TRUE))

# View the results
print(q1_3_results)
```

```
## # A tibble: 5 x 2
##   treatment                                       sd_ssm
```

```
##   <chr>                                          <dbl>
## 1 No Contact                                      1.68
## 2 Recycling Script by Gay Canvasser               1.70
## 3 Recycling Script by Straight Canvasser          1.68
## 4 Same-Sex Marriage Script by Gay Canvasser       1.67
## 5 Same-Sex Marriage Script by Straight Canvasser  1.68
```

```r
#ANSWER
# The standard deviations are very similar across all treatment groups (ranging from 1.67 to 1.70).
# This is important in a randomized experiment because it indicates that the groups are
# "balanced" not just in their average support, but also in their diversity of opinion.
# It suggests that the random assignment successfully distributed people with different
# baseline attitudes evenly across all experimental conditions.

# Q1.4:
# create the plot
ggplot(q1_2_results, aes(x = treatment, y = mean_ssm, fill = treatment)) +
  geom_col() +
  coord_flip() +
  labs(title = "Mean SSM Support by Treatment Group",
       subtitle = "Study 1, Wave 1 (Baseline)",
       x = "Treatment Condition",
       y = "Mean Support (1-5 scale)") +
  theme_minimal() +
  theme(legend.position = "none")
```



```r
#1.2 Checking Balance and Comparing Outcomes
#Q1.5
balance_check <- study1_wave1 %>%
  filter(treatment %in% c("Same-Sex Marriage Script by Gay Canvasser", "No Contact")) %>%
  group_by(treatment) %>%
  summarize(
    mean_ssm = mean(ssm, na.rm = TRUE),
    sd_ssm = sd(ssm, na.rm = TRUE),
    var_ssm = var(ssm, na.rm = TRUE) # Variance is s^2
  )
stats <- balance_check$mean_ssm
vars <- balance_check$var_ssm

std_diff <- (stats[2] - stats[1]) / sqrt((vars[2] + vars[1]) / 2)
print(std_diff)
```

```
## [1] -0.01049935
```

```r
# Answer - The standardized difference in means is -0.01, which is significantly
# lower than the 0.25 threshold. This indicates that the groups are
# very well-balanced at baseline. This is important
# because it ensures that any changes we see in later waves are
# likely due to the treatment (the canvassing) rather than
# pre-existing differences between the people in each group.

#Q1.6

study1_wave2 <- gay %>%
  filter(study == 1, wave == 2)
#calculate the mean ssm for the two groups
q1_6_results <- study1_wave2 %>%
  filter(treatment %in% c("Same-Sex Marriage Script by Gay Canvasser", "No Contact")) %>%
  group_by(treatment) %>%
  summarize(mean_ssm = mean(ssm, na.rm = TRUE))
print(q1_6_results)
```

```
## # A tibble: 2 x 2
##   treatment                                   mean_ssm
##   <chr>                                          <dbl>
## 1 No Contact                                      3.04
## 2 Same-Sex Marriage Script by Gay Canvasser       3.14
```

```r
diff_means_w2 <- q1_6_results$mean_ssm[2] - q1_6_results$mean_ssm[1]
print(diff_means_w2)
```

```
## [1] 0.09987463
```

```r
#Answer
# The difference in means at Wave 2 is approximately 0.10.
# Since we established in Q1.5 that these groups were balanced at baseline,
# this difference suggests that the 'same-Sex marriage Script' delivered
```

```r
# by a gay canvasser successfully increased support for same-sex marriage.
# Because the study used a randomized design, we can attribute this increase
# to the treatment effect of the conversation rather than pre-existing differences.

# Part 2 The Forensic Discovery
#Q.2.1
# observations and column names for gayreshaped.csv
nrow(gay_reshaped)
```

```
## [1] 11948
```

```r
colnames(gay_reshaped)
```

```
## [1] "study"     "treatment" "therm1"    "therm2"    "therm3"    "therm4"
```

```r
# observations and column names for ccap2012.csv
nrow(ccap)
```

```
## [1] 43998
```

```r
colnames(ccap)
```

```
## [1] "...1"     "caseid"   "gaytherm"
```

```r
# Answer
# gayreshaped.csv has 11,948 observations.
# Its columns are: "study", "treatment", "therm1", "therm2", "therm3", "therm4".
# ccap2012.csv has 43,998 observations.
# Its columns are: "...1" (index), "caseid", and "gaytherm".

#Q2.2
#Filter Study 1 from gayreshaped and calculate stats for therm1
gay_s1_stats <- gay_reshaped %>%
  filter(study == 1) %>%
  summarize(
    mean = mean(therm1, na.rm = TRUE),
    median = median(therm1, na.rm = TRUE),
    sd = sd(therm1, na.rm = TRUE)
  )

#calculate stats for gaytherm
ccap_stats <- ccap %>%
  summarize(
    mean = mean(gaytherm, na.rm = TRUE),
    median = median(gaytherm, na.rm = TRUE),
    sd = sd(gaytherm, na.rm = TRUE)
  )

#side by side
print("LaCour Study 1 (therm1) Statistics:")
```

```
## [1] "LaCour Study 1 (therm1) Statistics:"
```

```r
print(gay_s1_stats)
```

```
## # A tibble: 1 x 3
##    mean median    sd
##   <dbl>  <dbl> <dbl>
## 1  58.4     52  28.5
```

```r
print("CCAP 2012 (gaytherm) Statistics:")
```

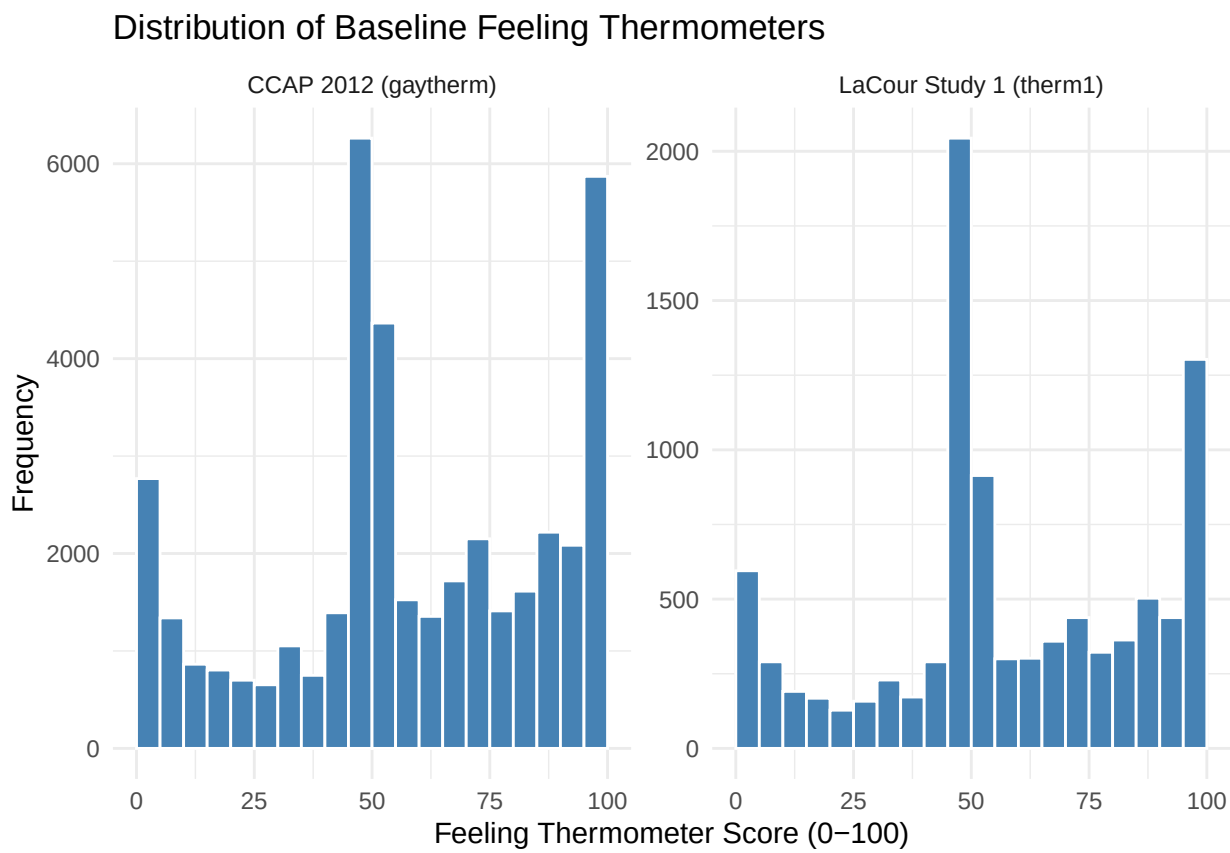
```
## [1] "CCAP 2012 (gaytherm) Statistics:"
```

```r
print(ccap_stats)
```

```
## # A tibble: 1 x 3
##    mean median    sd
##   <dbl>  <dbl> <dbl>
## 1  58.7     54  29.4
```

```r
# Answer - what i notice
# I notice that the mean (58.4 vs 58.7), median (52 vs 54), and standard
# deviation (28.5 vs 29.4) are almost identical between the LaCour Study 1
# and the CCAP 2012 survey. This is highly suspicious for a randomized
# experiment, as it suggests the experimental data was not collected
# independently but was likely "sampled" or copied from the existing
# CCAP distribution to appear realistic.

#Q2.3
#create a combined data frame for plotting
lacour_plot <- gay_reshaped %>%
  filter(study == 1) %>%
  select(therm = therm1) %>%
  mutate(source = "LaCour Study 1 (therm1)")
ccap_plot <- ccap %>%
  select(therm = gaytherm) %>%
  mutate(source = "CCAP 2012 (gaytherm)")
combined_data <- bind_rows(lacour_plot, ccap_plot)
#side-by-side histograms
ggplot(combined_data, aes(x = therm)) +
  geom_histogram(breaks = seq(0, 100, by = 5),
                 fill = "steelblue",
                 color = "white") +
  facet_wrap(~source, scales = "free_y") +
  labs(title = "Distribution of Baseline Feeling Thermometers",
       x = "Feeling Thermometer Score (0-100)",
       y = "Frequency") +
  theme_minimal()
```

```
## Warning: Removed 3097 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



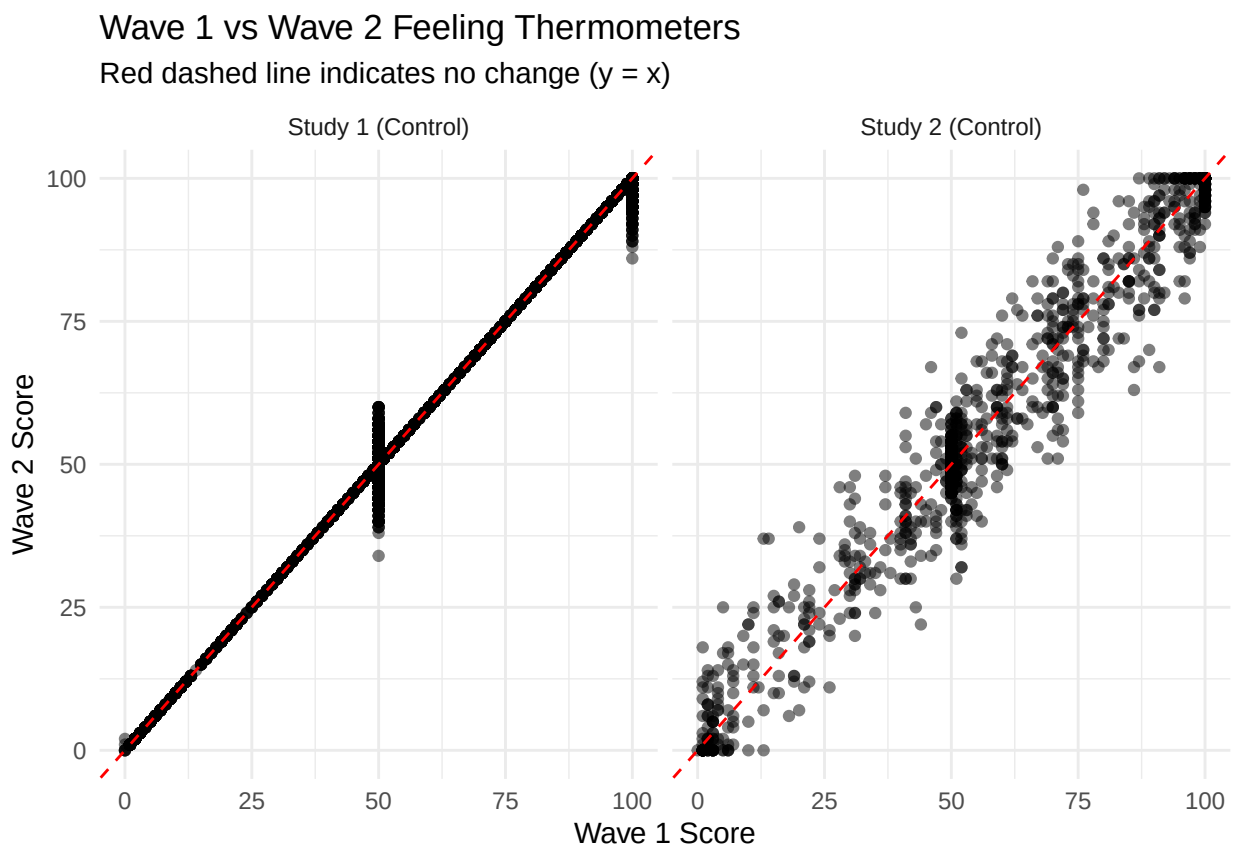

```
#Answer -
#These distributions do not look like they came from different populations.
#Despite being collected at different times from different groups, the
#histograms are nearly identical in shape and frequency of specific values.
# In a real-world experiment, we would expect more variation this
# level of similarity suggests the LaCour data was likely fabricated by
# copying the distribution of the existing CCAP survey.

#Q2.4
#no Contact groups in both studies
s1_control <- gay_reshaped %>% filter(study == 1, treatment == "No Contact")
s2_control <- gay_reshaped %>% filter(study == 2, treatment == "No Contact")

#clculate Correlations and SD of Change for Study 1
cor_s1 <- cor(s1_control$therm1, s1_control$therm2, use = "complete.obs")
sd_change_s1 <- sd(s1_control$therm2 - s1_control$therm1, na.rm = TRUE)

#correlations and SD of Change for Study 2
cor_s2 <- cor(s2_control$therm1, s2_control$therm2, use = "complete.obs")
sd_change_s2 <- sd(s2_control$therm2 - s2_control$therm1, na.rm = TRUE)

# side by side
results_2_4 <- tibble(
  Study = c("Study 1", "Study 2"),
  Correlation = c(cor_s1, cor_s2),
  SD_of_Change = c(sd_change_s1, sd_change_s2)
```

```r
)
print(results_2_4)
```

```
## # A tibble: 2 x 3
##   Study   Correlation SD_of_Change
##   <chr>         <dbl>        <dbl>
## 1 Study 1       0.998         1.99
## 2 Study 2       0.973         6.61
```

```r
#Answer
# Study 2 shows a more plausible pattern of re-test reliability with a correlation
# of 0.97 and an SD of change of 6.61.Study 1 shows a near-perfect
# correlation of 0.998 and a tiny SD of change (1.99). This suggests that the
# Wave 2 data in Study 1 was likely generated by taking Wave 1 and adding a very
# small amount of random noise, rather than actually re-surveying real people
# whose opinions naturally fluctuate more than that.

#Q2.5
#combine the control groups
combined_controls <- bind_rows(
  s1_control %>% mutate(label = "Study 1 (Control)"),
  s2_control %>% mutate(label = "Study 2 (Control)")
)

#side-by-side scatter plots
ggplot(combined_controls, aes(x = therm1, y = therm2)) +
  geom_point(alpha = 0.5) +
  # Add a 45-degree line (y = x)
  geom_abline(intercept = 0, slope = 1, color = "red", linetype = "dashed") +
  facet_wrap(~label) +
  labs(title = "Wave 1 vs Wave 2 Feeling Thermometers",
       subtitle = "Red dashed line indicates no change (y = x)",
       x = "Wave 1 Score",
       y = "Wave 2 Score") +
  theme_minimal()
```

```
## Warning: Removed 732 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
# Answer
#Study 2 looks much more like real data because the points form a
#distinct cloud around the 45-degree line, representing natural human
#inconsistency and minor shifts in opinion. Study 1 shows
#almost zero deviation from the 45-degree line. This lack of any noise or
#measurement error suggests the data was mathematically generated rather than
#collected from real people, as human respondents would never be this
#perfectly consistent over multiple weeks.

#2.3
# Q2.6
#The combination of these three irregularities provides evidence of
#data fabrication. First, the baseline "therm1" statistics (Mean: 58.4, SD: 28.5)
#almost perfectly match the independent CCAP 2012 survey (Mean: 58.7, SD: 29.4),
#which is statistically impossible for a new, independent sample. Second, the
#re-test correlation of 0.998 in Study 1 is far above the plausible human
#reliability range of 0.95-0.97. Third, the tiny SD of within-person change (1.99)
#shows a lack of natural human "noise" or rounding variation over time.
#A fabricator likely produced data with these properties because they
# wanted the results to look clean and authoritative by copying a
# real-world distribution (CCAP) and adding a minute amount of random
# noise, they created an illusion of high-quality data that could pass
# a surface-level check but failed under forensic scrutiny.

#Part 3
#Q3.1
```

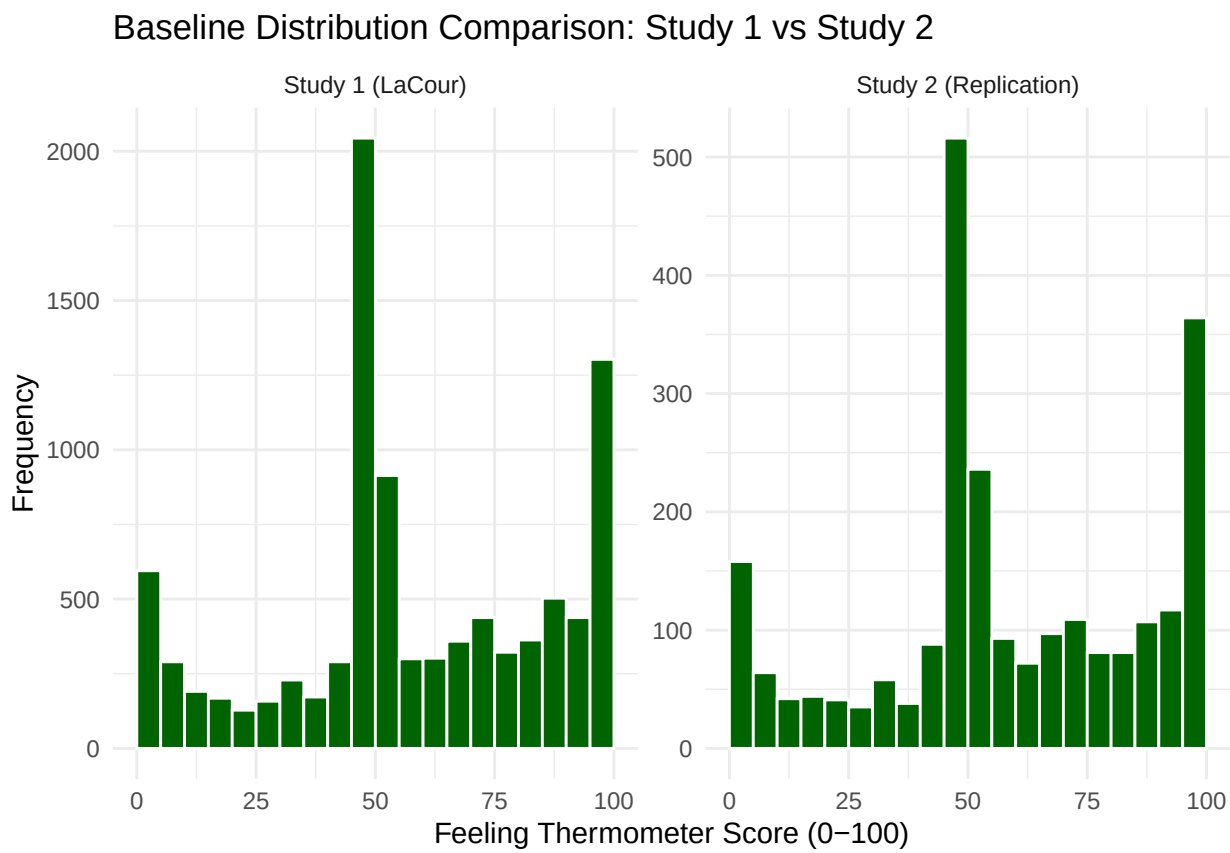
```r
study2_data <- gay_reshaped %>%
  filter(study == 2)
nrow(study2_data)
```

```
## [1] 2441
```

```r
study2_baseline_stats <- study2_data %>%
  group_by(treatment) %>%
  summarize(
    mean_therm1 = mean(therm1, na.rm = TRUE),
    sd_therm1 = sd(therm1, na.rm = TRUE),
    n = n()
  )

print(study2_baseline_stats)
```

```
## # A tibble: 2 x 4
##   treatment                                  mean_therm1 sd_therm1     n
##   <chr>                                            <dbl>     <dbl> <int>
## 1 No Contact                                        57.9      28.7  1203
## 2 Same-Sex Marriage Script by Gay Canvasser         59.4      28.4  1238
```

```r
#answer
#study 2 contains 2,441 observations. The baseline means for the
#"No Contact" group (57.9) and the "Gay Canvasser" group (59.4)
# are relatively close, suggesting that randomization successfully
# balanced the groups, though they are not suspiciously identical.

#Q3.2
s1_plot_data <- gay_reshaped %>%
  filter(study == 1) %>%
  select(therm = therm1) %>%
  mutate(source = "Study 1 (LaCour)")

s2_plot_data <- gay_reshaped %>%
  filter(study == 2) %>%
  select(therm = therm1) %>%
  mutate(source = "Study 2 (Replication)")

combined_studies <- bind_rows(s1_plot_data, s2_plot_data)
#histogramss
ggplot(combined_studies, aes(x = therm)) +
  geom_histogram(breaks = seq(0, 100, by = 5),
                 fill = "darkgreen",
                 color = "white") +
  facet_wrap(~source, scales = "free_y") +
  labs(title = "Baseline Distribution Comparison: Study 1 vs Study 2",
       x = "Feeling Thermometer Score (0-100)",
       y = "Frequency") +
  theme_minimal()
```




```
#answer
#Visually both Study 1 and 2 show "heaping" (spikes) at the
# values of 0, 50, and 100, which is typical for feeling thermometers
#where respondents round to neutral or extreme values.
#However, study 1 is a suspiciously perfect match to the national
# CCAP distribution. Study 2, while similar in its
# overall range, shows more natural "noise" and variation in the
# heights of these heaps, suggesting it is an independent sample
# rather than a copied distribution.

#3.2
# Q3.3
s2_balance_data <- gay_reshaped %>%
  filter(study == 2,
         treatment %in% c("No Contact", "Same-Sex Marriage Script by Gay Canvasser"))
#Mean, SD, and Variance for each group
s2_balance_stats <- s2_balance_data %>%
  group_by(treatment) %>%
  summarize(
    mean = mean(therm1, na.rm = TRUE),
    sd = sd(therm1, na.rm = TRUE),
    var = var(therm1, na.rm = TRUE)
  )

mean_tr_s2 <- s2_balance_stats$mean[2]
mean_co_s2 <- s2_balance_stats$mean[1]
```

```r
var_tr_s2  <- s2_balance_stats$var[2]
var_co_s2  <- s2_balance_stats$var[1]

std_diff_s2 <- (mean_tr_s2 - mean_co_s2) / sqrt((var_tr_s2 + var_co_s2) / 2)


print(s2_balance_stats)
```

```
## # A tibble: 2 x 4
##   treatment                                  mean    sd   var
##   <chr>                                     <dbl> <dbl> <dbl>
## 1 No Contact                                 57.9  28.7  823.
## 2 Same-Sex Marriage Script by Gay Canvasser  59.4  28.4  807.
```

```r
cat("Standardized Difference in Means (Study 2):", std_diff_s2, "\n")
```

```
## Standardized Difference in Means (Study 2): 0.05128243
```

```r
#answer
#The standardized difference in means for Study 2 is approximately 0.05,
#which suggests that the treatment and control groups are well-balanced
#at baseline. Unlike the too-perfect balance often seen in fabricated
#data, this small but non-zero difference is exactly what we expect
#from a legitimate randomization process in a real-world experiment.

#3.4
#mean of therm2 for each group
s2_post_stats <- s2_balance_data %>%
  group_by(treatment) %>%
  summarize(
    mean_therm2 = mean(therm2, na.rm = TRUE),
    n = n()
  )

#means to calculate the difference
mean_tr_post <- s2_post_stats$mean_therm2[2]
mean_co_post <- s2_post_stats$mean_therm2[1]
#ATE
ate_s2 <- mean_tr_post - mean_co_post
print(s2_post_stats)
```

```
## # A tibble: 2 x 3
##   treatment                                  mean_therm2     n
##   <chr>                                            <dbl> <int>
## 1 No Contact                                        57.6  1203
## 2 Same-Sex Marriage Script by Gay Canvasser         61.2  1238
```

```r
cat("Difference in post-period means (Study 2):", ate_s2, "\n")
```

```
## Difference in post-period means (Study 2): 3.567885
```

```r
# answer
#In Study 2, the difference in post-treatment means is 3.57 points.
#While this indicates a positive effect from the gay canvasser script,
# it is notably smaller than the effect size reported in Study 1. This
# suggests that the original study inflated the impact of
# the intervention, which is common when data is fabricated to produce
# more exciting or headline worthy results.

#Q3.5
#variance and n for each group
s2_uncertainty_stats <- s2_balance_data %>%
  group_by(treatment) %>%
  summarize(
    var_therm2 = var(therm2, na.rm = TRUE),
    n = n()
  )

#t values for the SE formula
var_tr <- s2_uncertainty_stats$var_therm2[2]
var_co <- s2_uncertainty_stats$var_therm2[1]
n_tr   <- s2_uncertainty_stats$n[2]
n_co   <- s2_uncertainty_stats$n[1]
# SE
se_diff <- sqrt((var_tr / n_tr) + (var_co / n_co))
lower_ci <- ate_s2 - (1.96 * se_diff)
upper_ci <- ate_s2 + (1.96 * se_diff)

cat("Standard Error:", se_diff, "\n")
```

```
## Standard Error: 1.160643
```

```r
cat("95% Confidence Interval: [", lower_ci, ",", upper_ci, "]\n")
```

```
## 95% Confidence Interval: [ 1.293024 , 5.842745 ]
```

```r
# answer
#The 95% confidence interval for the difference in post-treatment means
#is [1.29, 5.84]. Since this interval does not include zero, we can
#conclude that the difference is statistically significant at the 5% level
#This confirms that while the "real" effect in Study 2 (3.57)
#is much smaller than the fabricated 10-point effect in study 1, the
#gay canvasser intervention still produced a measurable and statistically
# reliable increase in support for same-sex marriage.
```